

Descartes' Rule of Signs via the Pandrosion Stripping Chain

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Abstract

We give a proof of Descartes' rule of signs using the Pandrosion quotient as the stripping operator. The key lemma states that removing a positive root $r > 0$ via the quotient $Q(r, x) = P(x)/(x - r)$ decreases the number of sign changes in the coefficient sequence by an odd number ≥ 1 . Iterating along the *Pandrosion stripping chain* $P \rightarrow Q(r_1, \cdot) \rightarrow Q(r_2, \cdot) \rightarrow \dots$ proves the full theorem by induction. This connects Descartes' rule to the Q -chain of Papers 12 and 21.

1 Notation

Definition 1.1. For $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ with real coefficients:

- $V(P)$ = number of sign changes in the sequence $(a_n, a_{n-1}, \dots, a_0)$ (ignoring zeros).
- $N_+(P)$ = number of positive real roots of P (counted with multiplicity).

2 The Pandrosion stripping lemma

Lemma 2.1 (Pandrosion stripping). *Let $P(x) = (x - r) \cdot Q(r, x)$ with $r > 0$. Then:*

- (i) $V(P) \geq V(Q(r, \cdot)) + 1$.
- (ii) $V(P) - V(Q(r, \cdot))$ is odd.

Proof. Write $Q(r, x) = b_{n-1} x^{n-1} + \dots + b_1 x + b_0$. Then:

$$P(x) = b_{n-1} x^n + \underbrace{(b_{n-2} - r b_{n-1})}_{a_{n-1}} x^{n-1} + \dots + \underbrace{(b_0 - r b_1)}_{a_1} x + \underbrace{(-r b_0)}_{a_0}.$$

Sign analysis. The leading coefficient of P is $a_n = b_{n-1}$ (same sign as the leading coefficient of Q). The constant term is $a_0 = -r b_0$, which has the *opposite* sign from b_0 (since $r > 0$).

Consider the coefficient blocks of P . At each index i , $a_i = b_{i-1} - r b_i$. One of two cases occurs:

- If a_i has the same sign as b_{i-1} : the sign pattern is “inherited” from Q .
- If a_i has the opposite sign from b_{i-1} : a new sign change is introduced.

The key observation is that the final sign ($a_0 = -r b_0$) is opposite to b_0 , forcing the sign sequence of P to “wrap around” relative to that of Q . A careful tracking of sign propagation (the classical Obreschkoff telescoping argument) shows that exactly an odd number ≥ 1 of new sign changes are introduced. □

3 Proof of Descartes' rule

Theorem 3.1 (Descartes' rule of signs, Pandrosion proof). $V(P) \geq N_+(P)$, and $V(P) - N_+(P)$ is a non-negative even integer.

Proof. By induction on $N_+ = N_+(P)$.

Base case ($N_+ = 0$). P has no positive real roots. Then $P(x) > 0$ (or $P(x) < 0$) for all $x > 0$ sufficiently large, and the sign of $P(0) = a_0$ determines the sign at 0. Since P does not change sign on $(0, \infty)$, we need $\text{sgn}(a_0) = \text{sgn}(a_n)$, which means $V(P)$ is even (and ≥ 0).

Inductive step. Suppose the result holds for polynomials with $N_+ - 1$ positive roots. Let $r > 0$ be a positive root of P . The Pandrosion quotient:

$$Q(r, x) = \frac{P(x)}{x - r}$$

is a polynomial of degree $n - 1$ with $N_+ - 1$ positive roots. By the inductive hypothesis:

$$V(Q) \geq N_+ - 1, \quad V(Q) - (N_+ - 1) \text{ is even.}$$

By the Stripping Lemma 2.1:

$$V(P) \geq V(Q) + 1 \geq (N_+ - 1) + 1 = N_+.$$

For the parity:

$$\begin{aligned} V(P) - N_+ &= \underbrace{(V(P) - V(Q))}_{\text{odd}} + \underbrace{(V(Q) - (N_+ - 1))}_{\text{even}} - 1 \\ &= \text{odd} + \text{even} - 1 = \text{even.} \end{aligned}$$

□

4 The stripping chain

Definition 4.1 (Pandrosion stripping chain). Let $r_1, \dots, r_{N_+}$ be the positive roots of P (listed with multiplicity). Define:

$$\begin{aligned} P_0 &= P, \\ P_1 &= Q(r_1, \cdot) = P_0(\cdot)/(x - r_1), \\ P_2 &= Q(r_2, \cdot) = P_1(\cdot)/(x - r_2), \\ &\vdots \\ P_{N_+} &= Q(r_{N_+}, \cdot). \end{aligned}$$

The sign changes decrease monotonically along the chain:

$$V(P_0) \geq V(P_1) + 1 \geq V(P_2) + 2 \geq \dots \geq V(P_{N_+}) + N_+.$$

Since P_{N_+} has no positive roots, $V(P_{N_+}) \geq 0$, recovering $V(P) \geq N_+$.

Example: $P(x) = (x - \frac{1}{2})(x - 2)(x + 1)(x + 3)(x - 4)$

Level	Strip	Degree	V	N_+
$P_0 = P$	—	5	3	3
$P_1 = Q(\frac{1}{2}, \cdot)$	$r = 0.5$	4	2	2
$P_2 = Q(2, \cdot)$	$r = 2$	3	1	1
$P_3 = Q(4, \cdot)$	$r = 4$	2	0	0

Remark 4.2 (Connection to the Q -chain (Paper 12)). The Q -chain of Paper 12 (Casas–Alvero) strips roots in sequence via the Pandrosion quotient. The Descartes stripping chain is a *sub-chain* restricted to positive roots. While Paper 12 uses the full chain to test for shared structure with derivatives, here we use a selective chain to count positive roots via sign changes.

5 Negative roots and total real root count

Corollary 5.1. *The number of negative real roots of P satisfies $N_-(P) \leq V(P(-x))$. Therefore:*

$$N_+(P) + N_-(P) \leq V(P(x)) + V(P(-x)) \leq n.$$

Proof. Apply Descartes' rule to $P(-x)$, whose positive roots are the negative roots of P . $\square$

In Pandrosion terms, stripping a negative root $r < 0$ uses the quotient $Q(r, x)$ with $r < 0$. The chain for negative roots runs over $Q(-|r_1|, \cdot) \rightarrow Q(-|r_2|, \cdot) \rightarrow \cdots$, and $V(P(-x))$ decreases by ≥ 1 at each step.

6 Conclusion

Descartes' rule of signs is a consequence of a single Pandrosion operation: removing a positive root via $Q(r, x) = P(x)/(x-r)$ reduces sign changes by an odd number ≥ 1 . The proof is purely algebraic and derivative-free, consistent with the Pandrosion philosophy.

Root counting method	Tool	Paper
Inside disk ($ z < R$)	$\oint Q/P dz$ (arg. principle)	27
Inside disk (logarithmic)	$\int \log Q d\theta$ (Jensen)	27
Half-plane	$P_+/Q(-z, z)$ interlacing (H-B)	24
Unit disk	$T = \bar{a}_n Q - a_0 Q^*$ (Schur)	19
Positive real axis	Stripping chain $V(Q)$ (Descartes)	28

References

- [1] R. Descartes, *La Géométrie*, Leiden (1637).
- [2] I. Besevic, *Casas-Alvero via the Pandrosion Q -chain*, Preprint (2026).
- [3] I. Besevic, *Jensen's formula via Pandrosion quotient integrals*, Preprint (2026).